

Text Latent Representation Using Tokenization and V

Customer review dataset laboratory report

Objective

Transform raw customer reviews into numerical representations using preprocessing, tokenization, padding, dense word embeddings, flattening, and PCA.

Dataset

- The product quality is excellent
- I really love this product
- The quality of this item is amazing
- The product performance is poor
- I dislike this product completely

Method summary

Lowercase and clean text → assign token IDs → pad to a common length → map IDs to four-dimensional vectors → flatten each matrix → project word vectors with PCA.

Note

The embeddings in this report are reproducible illustrative vectors generated with a fixed random seed. They are not pretrained semantic embeddings.

Vocabulary and Token Sequences

Vocabulary with corresponding token IDs

Word	Token ID	Word	Token ID
the	1	product	2
quality	3	is	4
excellent	5	i	6
really	7	love	8
this	9	of	10
item	11	amazing	12
performance	13	poor	14
dislike	15	completely	16

Numerical representation and padding to length 7

Sample	Token sequence	Padded sequence
1	[1, 2, 3, 4, 5]	[1, 2, 3, 4, 5, 0, 0]
2	[6, 7, 8, 9, 2]	[6, 7, 8, 9, 2, 0, 0]
3	[1, 3, 10, 9, 11, 4, 12]	[1, 3, 10, 9, 11, 4, 12]
4	[1, 2, 13, 4, 14]	[1, 2, 13, 4, 14, 0, 0]
5	[6, 15, 9, 2, 16]	[6, 15, 9, 2, 16, 0, 0]

Word Embedding Representation Sample 1

The product quality is excellent

Each token is represented by a dense vector with four dimensions. The final row is padding when the sentence is shorter than the maximum length.

Position	d1	d2	d3	d4
Position 1	0.001	0.299	-0.274	-0.891
Position 2	-0.455	-0.992	0.060	1.340
Position 3	-0.492	-0.620	0.490	0.357
Position 4	0.105	-0.930	-0.029	0.695
Position 5	-1.344	-0.458	-1.901	-1.290
Position 6	0.000	0.000	0.000	0.000
Position 7	0.000	0.000	0.000	0.000

Word Embedding Representation Sample 2

I really love this product

Each token is represented by a dense vector with four dimensions. The final row is padding when the sentence is shorter than the maximum length.

Position	d1	d2	d3	d4
Position 1	-1.842	-0.235	-1.267	0.271
Position 2	0.157	-0.187	-2.517	-0.539
Position 3	-0.049	0.113	-1.530	-0.478
Position 4	-0.979	-0.809	1.061	-0.808
Position 5	-0.455	-0.992	0.060	1.340
Position 6	0.000	0.000	0.000	0.000
Position 7	0.000	0.000	0.000	0.000

Word Embedding Representation Sample 3

The quality of this item is amazing

Each token is represented by a dense vector with four dimensions. The final row is padding when the sentence is shorter than the maximum length.

Position	d1	d2	d3	d4
Position 1	0.001	0.299	-0.274	-0.891
Position 2	-0.492	-0.620	0.490	0.357
Position 3	-0.033	0.884	-0.584	-0.112
Position 4	-0.979	-0.809	1.061	-0.808
Position 5	0.110	0.064	-1.225	0.076
Position 6	0.105	-0.930	-0.029	0.695
Position 7	1.359	-1.547	0.859	0.119

Word Embedding Representation Sample 4

The product performance is poor

Each token is represented by a dense vector with four dimensions. The final row is padding when the sentence is shorter than the maximum length.

Position	d1	d2	d3	d4
Position 1	0.001	0.299	-0.274	-0.891
Position 2	-0.455	-0.992	0.060	1.340
Position 3	-0.641	2.000	0.762	-1.199
Position 4	0.105	-0.930	-0.029	0.695
Position 5	0.075	0.577	-0.189	0.683
Position 6	0.000	0.000	0.000	0.000
Position 7	0.000	0.000	0.000	0.000

Word Embedding Representation Sample 5

I dislike this product completely

Each token is represented by a dense vector with four dimensions. The final row is padding when the sentence is shorter than the maximum length.

Position	d1	d2	d3	d4
Position 1	-1.842	-0.235	-1.267	0.271
Position 2	-0.067	0.667	1.439	-0.676
Position 3	-0.979	-0.809	1.061	-0.808
Position 4	-0.455	-0.992	0.060	1.340
Position 5	0.203	-0.463	0.127	-1.187
Position 6	0.000	0.000	0.000	0.000
Position 7	0.000	0.000	0.000	0.000

Flattened Vector Representations

Each 7×4 embedding matrix is flattened row by row into a single vector of length 28.

Sample 1

[0.001, 0.299, -0.274, -0.891, -0.455, -0.992, 0.060, 1.340, -0.492, -0.620, 0.490, 0.357, 0.105, -0.930, -0.029, 0.695, -1.344, -0.458, -1.901, -1.290, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000]

Sample 2

[-1.842, -0.235, -1.267, 0.271, 0.157, -0.187, -2.517, -0.539, -0.049, 0.113, -1.530, -0.478, -0.979, -0.809, 1.061, -0.808, -0.455, -0.992, 0.060, 1.340, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000]

Sample 3

[0.001, 0.299, -0.274, -0.891, -0.492, -0.620, 0.490, 0.357, -0.033, 0.884, -0.584, -0.112, -0.979, -0.809, 1.061, -0.808, 0.110, 0.064, -1.225, 0.076, 0.105, -0.930, -0.029, 0.695, 1.359, -1.547, 0.859, 0.119]

Sample 4

[0.001, 0.299, -0.274, -0.891, -0.455, -0.992, 0.060, 1.340, -0.641, 2.000, 0.762, -1.199, 0.105, -0.930, -0.029, 0.695, 0.075, 0.577, -0.189, 0.683, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000]

Sample 5

[-1.842, -0.235, -1.267, 0.271, -0.067, 0.667, 1.439, -0.676, -0.979, -0.809, 1.061, -0.808, -0.455, -0.992, 0.060, 1.340, 0.203, -0.463, 0.127, -1.187, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000, 0.000]

PCA Visualization and Analysis

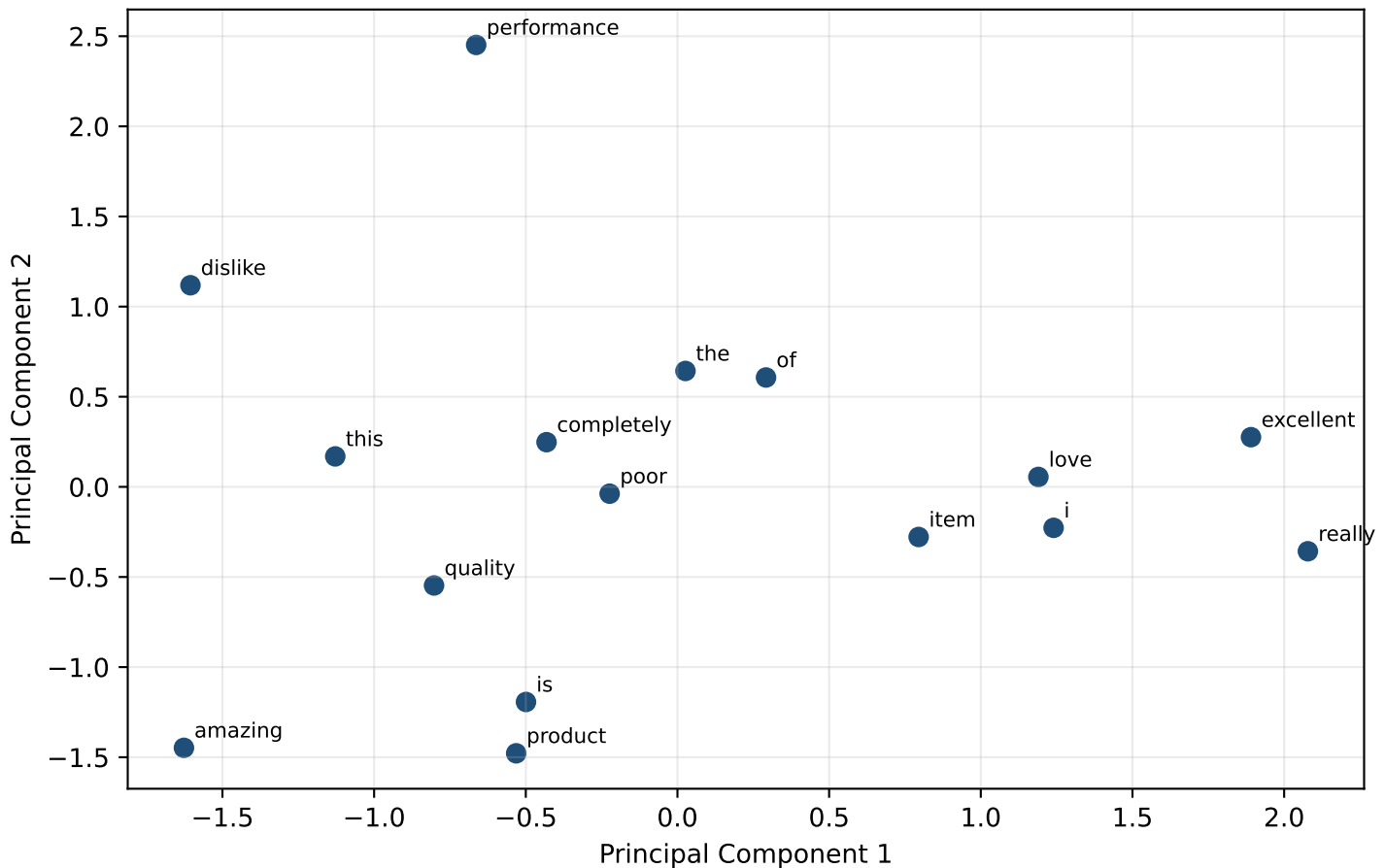


Figure 1. Two-dimensional PCA projection of the word embeddings.

Analysis

Tokenization converts the reviews into discrete IDs, while padding makes every sample compatible with a single rectangular tensor. The embedding layer changes each ID into a dense vector, preserving a fixed four-value representation per token. Flattening then produces a fixed-length sentence vector with 28 values.

Interpretation

PCA provides a compact visual summary of the embedding space. Because these vectors are illustrative and randomly generated, the plotted distances should not be interpreted as learned language semantics.